Chapter 2.3 — Formula Sheet

Order Statistics & Non-Parametric Extremes

Companion quick-reference to your Notebook 2 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

Distributions of Extremes

Sample Maximum & Minimum [Ex 1–4]

For $X_1, \ldots, X_n$ i.i.d. with CDF F, PDF f:

$$\mathbb{P}(\max \le x) = [F(x)]^n, \qquad f_{\max}(x) = n[F(x)]^{n-1} f(x)$$

$$\mathbb{P}(\min \ge x) = [1 - F(x)]^n, \qquad f_{\min}(x) = n[1 - F(x)]^{n-1} f(x)$$

Uniform special case: $X_{(n)} \sim \text{Beta}(n, 1)$, $X_{(1)} \sim \text{Beta}(1, n)$.
Exponential special case: $\min(X_1, \ldots, X_n) \sim \text{Exp}(n\lambda)$ for $X_i \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$ — matches the racing-Exponentials result (Ch. 1.3).

Joint Density of Min & Max, and the Range [Ex 5–6]

For $a < b$:

$$f_{(1),(n)}(a, b) = n(n-1)[F(b) - F(a)]^{n-2} f(a) f(b)$$

Speed trick: $\mathbb{P}(\min > a, \max < b)$ is just $[F(b) - F(a)]^n$ (all n values land in (a, b)) — skip the joint density unless the two bounds are asymmetric constraints.
Range $R = \max - \min$ for n i.i.d. Unif$(0, 1)$:

$$f_R(r) = n(n-1) r^{n-2}(1 - r), \quad \mathbb{E}[R] = \frac{n-1}{n+1}$$

General k-th Order Statistic [Ex 7–10]

$$f_{(k)}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

For Unif$(0, 1)$: $X_{(k)} \sim \text{Beta}(k,\ n - k + 1)$.
$\mathbb{E}[\max]$ **speed trick (no differentiation):** for non-negative bounded X,

$$\mathbb{E}[X] = \int_0^1 \mathbb{P}(X > x)\, dx \ \Rightarrow\ \mathbb{E}[\max] = \frac{n}{n+1} \ (\text{Unif}(0, 1))$$

Auction application: n bidders, valuations i.i.d. Unif$(0, 1)$. Winning value $= \mathbb{E}[X_{(n)}] = \frac{n}{n+1}$; second-price (Vickrey) payment $= \mathbb{E}[X_{(n-1)}] = \frac{n-1}{n+1}$.

Uniform Spacings

Setup & Symmetry [Ex 11]

n points break $[0, 1]$ into $n + 1$ spacings $S_1, \ldots, S_{n+1}$:

$$\sum_{i=1}^{n+1} S_i = 1$$

By rotational symmetry (points on a circle), every S_i has the **same marginal distribution**: $S_i \sim \text{Beta}(1, n)$.

Order-Statistic Mean & Variance [Ex 12–14]

Elegant derivation (no integration): $X_{(k)} = S_1 + \cdots + S_k$, and $\mathbb{E}[S_i] = \frac{1}{n+1}$ for every i, so by linearity:

$$\mathbb{E}[X_{(k)}] = \frac{k}{n+1}$$

Confirmed via Beta mean: $X_{(k)} \sim \text{Beta}(k, n - k + 1) \Rightarrow \mathbb{E} = \frac{k}{k+(n-k+1)} = \frac{k}{n+1}$.
Variance:

$$\text{Var}(X_{(k)}) = \frac{k(n-k+1)}{(n+1)^2(n+2)}$$

Rényi's Exponential Representation [Ex 15]

For $E_1, \ldots, E_{n+1} \overset{\text{iid}}{\sim} \text{Exp}(1)$:

$$(S_1, \ldots, S_{n+1}) \overset{d}{=} \left(\frac{E_1}{\sum_j E_j}, \ldots, \frac{E_{n+1}}{\sum_j E_j} \right)$$

Converts hard "broken stick" probability questions into Gamma/Exponential calculations.

"All Spacings $\le c$" & Covariance [Ex 16–18]

$$\mathbb{P}\left(\max_i S_i \le c \right) = \sum_{j=0}^{\lfloor 1/c \rfloor} (-1)^j \binom{n+1}{j} (1 - jc)_+^n$$

Special case $n = 2$, $c = \frac{1}{2}$**:** gives $\mathbb{P} = \frac{1}{4}$ — exactly the triangle-formation probability (Ch. 1.3), since "all 3 pieces $\le \frac{1}{2}$" *is* the triangle inequality.
Covariance (sum-is-constant trick): since $\sum S_i = 1$ is constant, $\text{Var}(\sum S_i) = 0$ gives

$$\text{Cov}(S_i, S_j) = -\frac{1}{(n+1)^2(n+2)} \quad (i \ne j)$$

Capstone: $\mathbb{E}[\text{Largest Spacing}]$ [Ex 19–20]

Rescaled spacing mean (e.g. order-arrival gaps over session length T): $\mathbb{E}[\text{gap}] = \frac{T}{n+1}$.

Exact closed form for the max spacing, via Rényi's representation and Gamma–Dirichlet independence:

$$\mathbb{E}\left[\max_i S_i \right] = \frac{\mathbb{E}[\text{max of } n{+}1 \text{ iid Exp}(1)]}{\mathbb{E}[T]} = \boxed{\frac{H_{n+1}}{n+1}}$$

where $T = \sum_j E_j \sim \text{Gamma}(n+1, 1)$ and $H_{n+1} = \sum_{j=1}^{n+1} \frac{1}{j}$.
At $n = 1$: $H_2/2 = 0.75$, matching $\max(X, 1 - X) \sim \text{Unif}(0.5, 1)$ from Ch. 1.3.

One-tool-tree reminder: every "hard" formula here (joint density, k-th order density, broken-stick inclusion–exclusion) has a faster shortcut sitting next to it (match-the-interval trick, spacing-sum linearity, sum-is-constant covariance, Rényi representation). In competition, reach for the shortcut first — fall back to the general formula only when the shortcut doesn't fit the constraint shape.